

Supporting Information

AlloFusion: Allosteric Site Prediction Based on Language Models and Multi-Feature Fusion

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Table S1. The independent test set D24 of 24 allosteric proteins

No.	Protein Name	PDB ID	Chain	Residue Num.
1	3',5'-cyclic nucleotide phosphodiesterase 2A	1MC0	A	369
2	Adenylate cyclase type 10	4OYA	A	470
3	Aspartokinase 1, chloroplastic	2CDQ	A	510
4	BirA(Pyrococcus horikoshii)	1WQW	A	235
5	Cathepsin K	5J94	A	223
6	Cystathionine beta-synthase	4PCU	A	549
7	Dihydropteroate synthase	4NHV	A	297
8	Inosine-5'-monophosphate dehydrogenase	4DQW	A	509
9	LIM domain kinase 2	4TPT	A	303
10	Neurolysin, mitochondrial	4FXV	P	693
11	Heterodimeric HIF-2	6E3U	B	642
12	Chorismate mutase 1, chloroplastic	4PPV	A	280
13	NAD-dependent protein deacetylase sirtuin-1	5BTR	A	397
14	Threonine synthase 1, chloroplastic	2C2B	A	486

15	Ubiquitin-conjugating enzyme E2 D1	4QPL	A	155
16	Serine/threonine-protein kinase PAK 1	4ZLO	A	298
17	Metabotropic glutamate receptor 5, Endolysin, Metabotropic glutamate receptor 5	5CGC	A	444
18	Phosphoenolpyruvate carboxykinase, cytosolic [GTP]	4YW8	A	624
19	Isocitrate dehydrogenase [NADP] cytoplasmic	5DE1	A	413
20	Nuclear receptor ROR-gamma	5C4T	A	241
21	Thyroid hormone receptor alpha	3UVV	B	244
22	GTP pyrophosphokinase YjbM	5DED	A	218
23	DynE7	2XEM	B	150
24	TmaDAH7PS	3PG9	B	338

Note: To assess potential data redundancy, we performed pairwise sequence similarity analysis between D24 and the training dataset using MMseqs2. Most proteins in D24 exhibit less than 30% sequence similarity with any sequence in the training set. Only three notable cases were observed: 3PG9 vs 4RGS (74.1% similarity), 4ZLO vs 4AW0 (30.9% similarity), and 5DE1 vs 4JA8 (70.2% similarity).

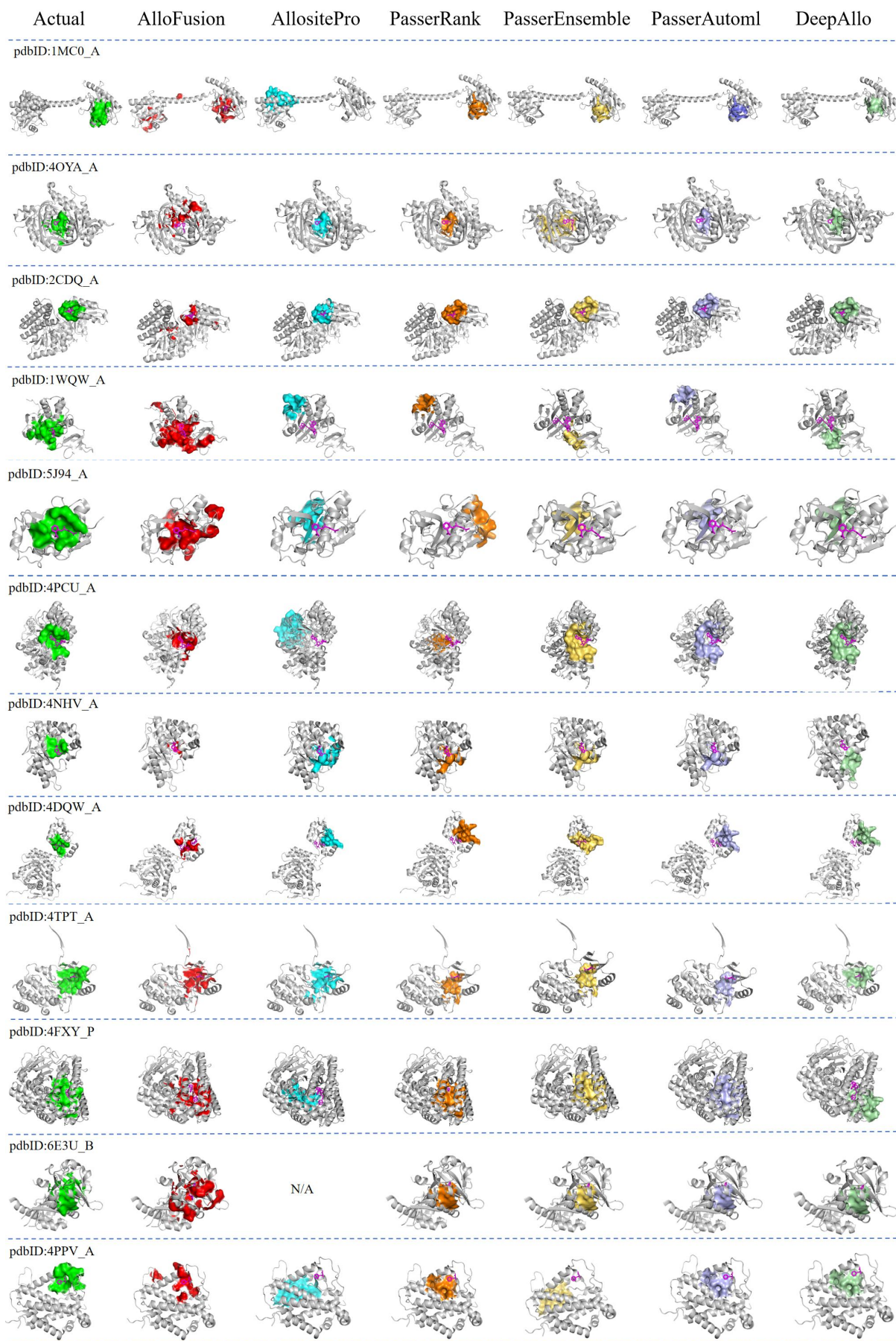
Table S2. Fine-Tuning Details of ProtT5

Parameters	Settings	Notes
optimizer	AdamW	Default parameters unless otherwise specified
learning rate	3e-4	Selected via prior grid search
batch size	4	Training batch size
Epochs	10	No early stopping, all epochs completed
lora_rank	4	Reduces the number of trainable parameters to ensure a balance between performance and computational efficiency.
lora_init_scale	0.01	Small-scale initialization prevents gradient explosion and maintains training stability.
lora_modules	.*SelfAttention . *EncDec Attention	Focuses fine-tuning on attention layers to better capture long-range dependencies relevant to allosteric site recognition.
lora_layers	q k v o	Covers all components of the attention mechanism, enhancing the model's ability to learn both local and global information.
trainable_param_names	.*layer_norm.* . *lora_[ab].*	Restricts fine-tuning to LoRA and LayerNorm layers, reducing computational overhead.
lora_scaling_rank	1	Maintains stable parameter updates and accommodates small-scale LoRA adjustments.

The fine-tuning of the ProtT5 model was conducted on an NVIDIA RTX 4090 GPU using the Hugging Face Trainer API. The following training hyperparameters were used: Learning rate: 3e-4, Training batch size: 4, Gradient accumulation steps: 2, Validation batch size: 4, Total epochs: 10, Evaluation strategy: Every 500 steps on the validation set, Precision: Mixed precision training (FP16), Optimizer: AdamW, Learning rate scheduler: Linear decay. No early stopping was applied. After completing all 10 training epochs, the model from the final epoch was selected as the final version.

Table S3. Hyperparameters used in Bayesian optimization, including search space and final selected values.

Hyperparameter	Description	Search Space	Final Value
conv1_filters	Filters in 1st Conv1D layer	[32, 64, 96, 128]	32
conv1_kernel_size	Kernel size in 1st Conv1D layer	[3, 5, 7]	3
dropout1_rate	Dropout rate after 1st Conv1D	[0.2, 0.5], step=0.1	0.2
conv2_filters	Filters in 2nd Conv1D layer	[64, 128, 192, 256]	128
conv2_kernel_size	Kernel size in 2nd Conv1D layer	[3, 5, 7]	3
dropout2_rate	Dropout rate after 2nd Conv1D	[0.2, 0.5], step=0.1	0.3
conv3_filters	Filters in 3rd Conv1D layer	[32, 64, 96, 128]	32
conv3_kernel_size	Kernel size in 3rd Conv1D layer	[3,5,7]	5
dropout3_rate	Dropout rate after 3rd Conv1D	[0.2, 0.5], step=0.1	0.2
conv4_filters	Filters in 4th Conv1D layer	[32, 64, 96, 128]	32
conv4_kernel_size	Kernel size in 4th Conv1D layer	[3,5,7]	3
dropout4_rate	Dropout rate after 4th Conv1D	[0.2, 0.5], step=0.1	0.3
dense_units	Units in fully connected layer	{64, 128, 192, 256}	128
learning_rate	Learning rate for Adam optimizer	[1e-5, 1e-2]	1e-5
Batch_size	Training batch size	fixed	32
epochs	Number of training epochs per trial	fixed	100



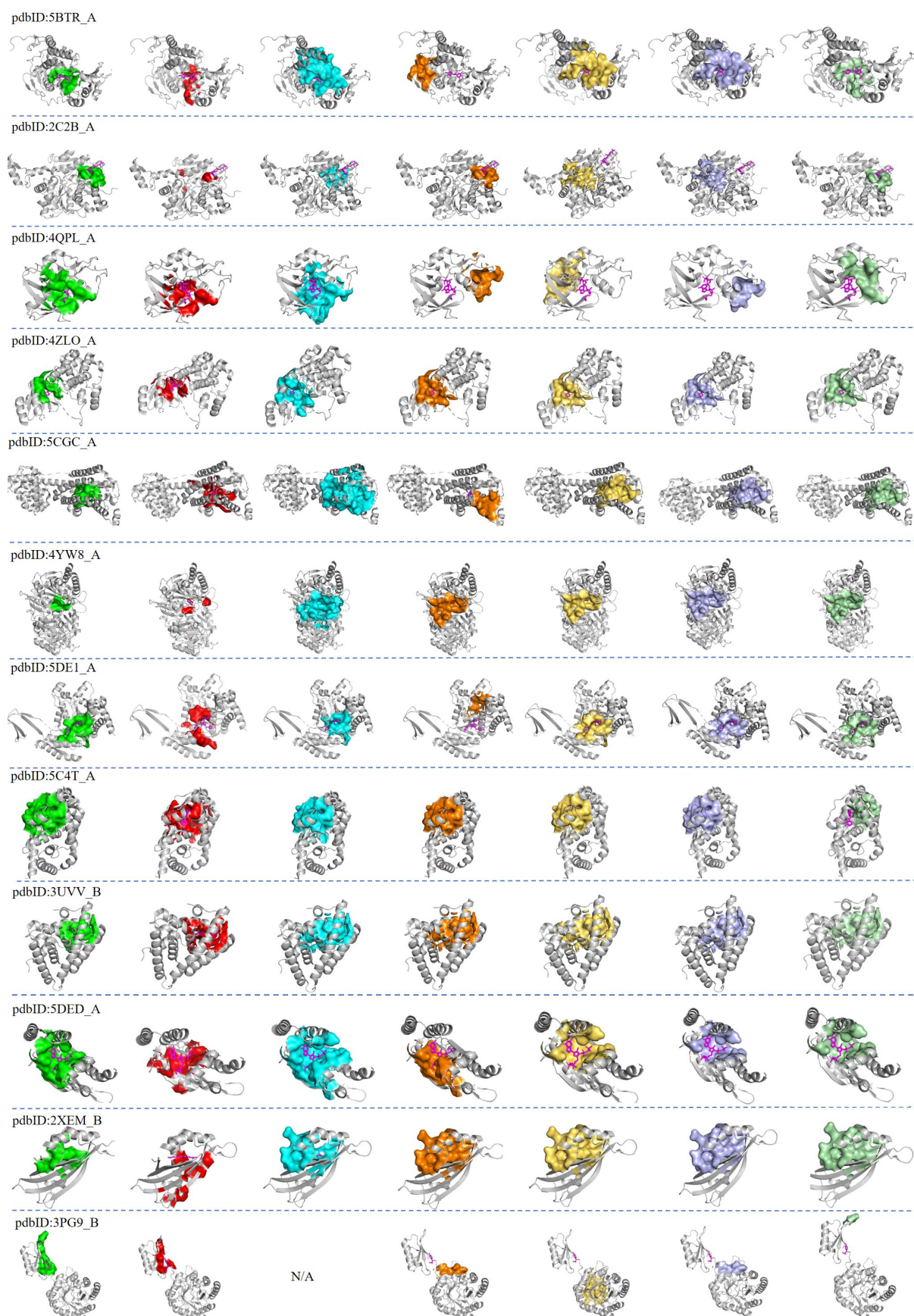


Figure S1. 3D representations of protein AFRs predicted by different models on the D24 dataset. Each row, from left to right, displays the experimentally determined AFRs, followed by predictions from

AlloFusion, AllositePro, PASSer Rank, PASSer Ensemble, PASSer Automl and DeepAllo. Green indicates the actual allosteric sites composed of AFRs; red, cyan, orange, light yellow, light blue and light green represent the predicted AFRs by AlloFusion, AllositePro, PASSer Rank, PASSer Ensemble, PASSer Automl and DeepAllo, respectively. Gray represents the cartoon structure of the protein. “N/A” denotes cases where no allosteric site is predicted by the corresponding method.

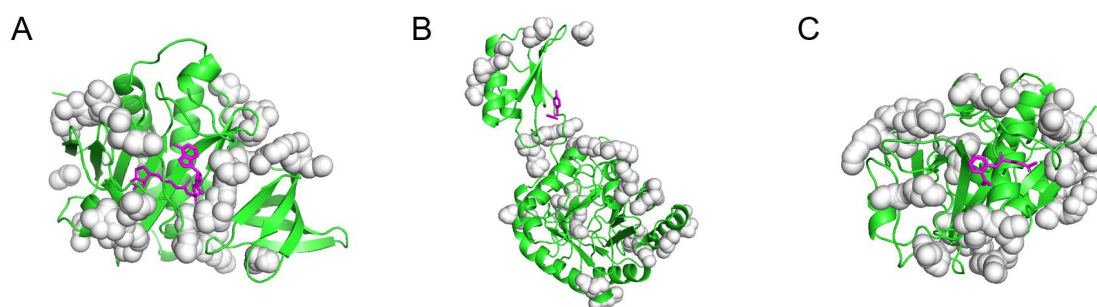


Figure S2. Pocket predictions by Fpocket on proteins (A) PDB ID: 1WQW, (B) PDB ID: 3PG9 and PDB ID: 5J94. Allosteric modulators (purple sticks) are not located within the predicted pockets (white spheres), indicating that these modulators may bind to non-pocket regions.

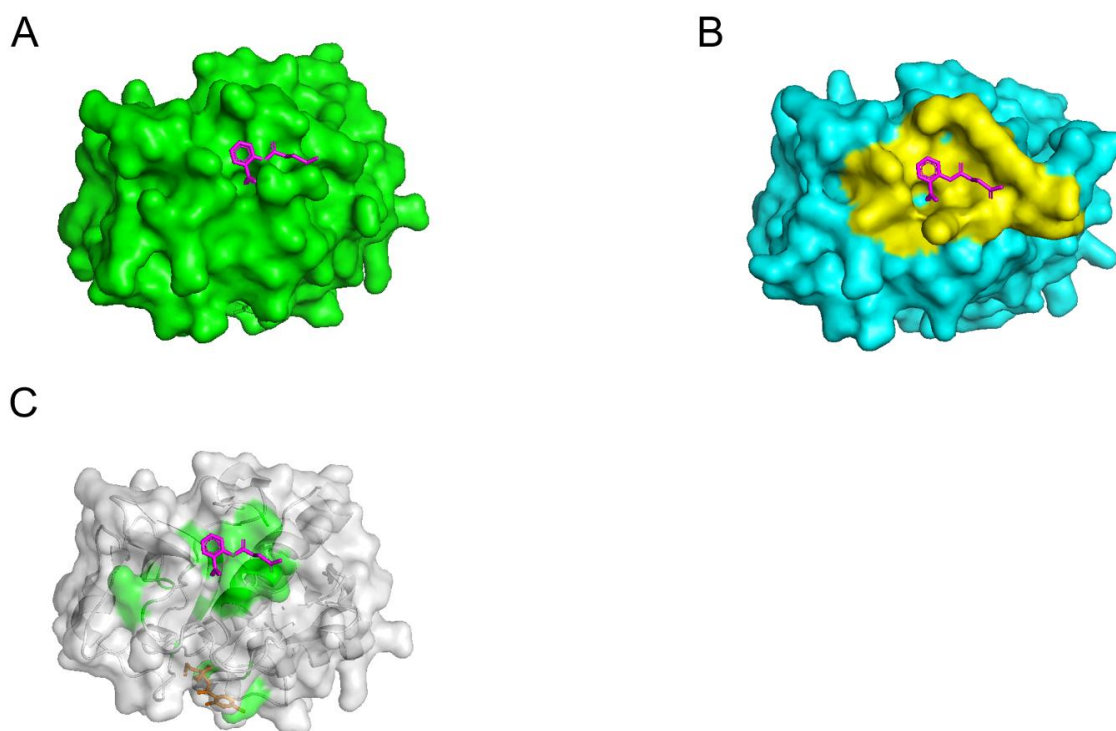


Figure S3. Structural comparison of PDB ID: 4X6I (ligand-free, apo state) and PDB ID: 5J94 (ligand-bound, holo state) illustrating the conformational behavior of the allosteric site. In the apo structure (Figure A, 4X6I), the putative allosteric site is occluded by surrounding structural elements. The modulator is manually added to the figure to highlight that this region is not accessible in the absence of the ligand. In the holo structure (Figure B, 5J94), conformational rearrangement exposes this region, allowing modulator binding. This transition is consistent with the characteristics of a cryptic allosteric site. Figure C shows the allosteric site predicted by AlloFusion on 4X6I.

Table S4. Distances between the predicted allosteric site centroids and the actual binding site on D24 using different methods.

PDB_Code	Chain	AlloFusion	AllositePro	PASSer_rank	PASSer Ensemble	PASSer_Automl	DeepAllo
1MC0	A	1.78	63.37	2.31	2.31	2.31	4.78
1WQW	A	2.6	15.9	24.18	16.53	24.18	12.69
2C2B	A	15.23	27.56	0.43	26.75	26.75	0.43
2CDQ	A	9.97	4.52	3.37	3.37	3.37	3.37
2XEM	B	8.25	2.31	2.83	2.83	2.83	2.83
3PG9	B	2.59	N/A	22.74	45.23	22.74	14.39
3UVV	B	1.48	2.58	2.43	2.43	2.43	2.43
4DQW	A	4.25	9.32	9.21	3.13	9.21	9.21
4FXV	P	6.72	31.97	1.01	1.01	1.01	15.97
4NHV	A	1.71	14.3	13.37	13.37	13.37	23.69
4OYA	A	6.96	2.81	24.79	21.81	24.79	2.43
4PCU	A	10.27	54.18	38.69	3.41	3.41	3.41
4PPV	A	13.03	22.31	12.87	23.95	12.87	12.87
4QPL	A	2.98	8.78	19.16	13.96	19.16	8.78
4TPT	A	3.11	1.72	4.58	4.58	4.58	4.68
4YW8	A	13.73	8.04	54.39	54.39	54.39	6.99
4ZLO	A	5.78	2.69	2.28	2.28	2.28	2.28
5BTR	A	12.59	6.23	13.22	3.19	3.19	13.22
5C4T	A	8.15	0.68	0.68	0.68	0.68	14.57
5CGC	A	4.53	4.5	16.31	4.77	4.77	4.77
5DE1	A	13.40	2.35	16.26	2.83	2.83	2.83
5DED	A	5.69	4.16	6.2	3.61	3.61	3.61
5J94	A	10.96	26.64	27.27	23.12	23.12	23.12
6E3U	B	6.15	N/A	2.77	2.77	2.77	2.77

Table S5. Number and false positive rate of AFRs for each protein in the D24 dataset.

PDB_Code	#Total Residues	#Actual AFR	#Predicted AFR	#True Positives (TP)	#False Positives (FP)	#True Negatives (TN)	FPR
1MC0_A	369	27	29	14	15	227	6.20%
1WQW_A	235	28	33	23	10	197	4.83%
2C2B_A	486	11	4	2	2	473	0.42%
2CDQ_A	510	12	7	4	3	495	0.60%
2XEM_B	150	13	13	2	11	190	5.47%
3PG9_B	338	15	9	8	1	322	0.31%
3UVV_B	244	18	17	11	6	220	2.65%
4DQW_A	509	14	9	6	3	492	0.61%
4FX_Y_P	693	21	18	15	3	669	0.45%
4NHV_A	297	4	8	4	4	289	1.37%
4OYA_A	470	22	15	10	5	443	1.11%
4PCU_A	549	15	11	6	5	529	0.97%
4PPV_A	280	9	12	4	8	263	2.95%
4QPL_A	155	17	15	12	3	135	2.22%
4TPT_A	303	20	21	15	6	277	2.12%
4YW8_A	624	10	12	4	8	605	1.31%
4ZLO_A	298	14	15	12	3	281	1.06%
5BTR_A	397	10	9	0	9	378	2.33%
5C4T_A	241	26	22	12	10	205	4.65%
5CGC_A	444	20	16	12	4	420	0.94%
5DE1_A	413	19	15	6	9	385	2.28%
5DED_A	218	21	14	7	7	188	3.59%
5J94_A	223	11	13	6	7	205	3.30%
6E3U_B	542	23	24	13	11	508	2.11%

$$* \text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$